

PHYSICAL / CHEMICAL CHANGES

Definitions

Physical Change: where a substance changes form or state with no new substance being formed.

e.g. Creating sawdust by sanding a piece of wood
- it is still wood but in smaller pieces.

Dissolving sugar into a cup of tea or coffee -
the sugar is still present in the solution, it
hasn't changed into another substance.

Melting of ice to form liquid water - you have
water in both phases.

Chemical Change: where a substance undergoes a chemical
reaction to form a new substance.

e.g. Iron will rust slowly in the atmosphere to form
rust (iron oxide).

Toast is burned bread - the new substance is the
carbon, which coats the bread.

Steak is cooked on a barbecue - the new
substance is carbon once again.

Generally physical changes are relatively easy to reverse since
they do not involve the formation of new chemical bonds.

e.g. Melting ice is easily frozen again in a freezer by the removal of heat.

Chemical changes are almost impossible to reverse since new chemical bonds are formed.

e.g. It would be impossible to change toast back into the slice of bread it was originally.

CHEMICAL REACTIONS

A chemical reaction has occurred if a new substance is formed.

Evidence that a reaction has occurred may be as follows.

1. A new colour is produced or some colour change is noted.

e.g. Mixing solutions of $\text{Pb}(\text{NO}_3)_2$ (lead nitrate) and NaI sodium iodide) produces a bright yellow solid (called a *precipitate*).

Colour change is not the most reliable test!

e.g. Heating blue CuSO_4 (copper sulphate) turns it white by driving off the water trapped in the crystal. However the CuSO_4 is still there unchanged.

2. A solid is produced.

e.g. The yellow solid produced in the example above.

3. A solid disappears.

e.g. Mg (magnesium) disappears when placed into HCl solution.

4. The reaction vessel goes warm (produces heat) or cold (absorbs heat).

e.g. Mg in HCl (hydrochloric acid) produces heat and the reaction vessel feels warm.

Ba(OH)₂ (barium hydroxide) added to NH₄SCN (ammonium thiocyanate) goes very cold as heat is absorbed from the surroundings to make the reaction “go”.

5. A gas is given off - detected by seeing bubbles or smelling something.

e.g. Mg in HCl produces bubbles of H₂ (hydrogen) gas in the solution.

FeS (iron sulphide) added to HCl produces no bubbles but a “rotten egg” gas (H₂S - hydrogen sulphide).